Antoine Kassis

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Education

- **M2 MVA (Mathématiques, Vision, Apprentissage)**, ENS Paris-Saclay 2026–2027
- **M1 Mathematics Magistère**, fundamental mathematics, Université Paris-Saclay 2025–2026
- **L3 Double Degree, Mathematics Magistère**, Université Paris-Saclay — *mention très bien* 2024–2025
- **L1–L2 Double Degree**, Mathematics with Physics then Computer Science, Université Paris-Saclay 2022–2024
- **French Baccalaureate**, Antonine International School, Lebanon — Highest Honors 2021–2022

Research

- **Research internship (TER)**, Institut Mathématique d'Orsay / INRIA DataShape — Marc Glisse, 19/20 Apr–Jun 2026
  Generalized Cohen-Steiner's curvature inequalities from curves and surfaces to arbitrary dimension. The main theorem bounds differences in Lipschitz–Killing curvatures by integrating the Euler characteristic over hyperplane sections. Persistent homology, integral geometry (Crofton, Steiner, Hadwiger), Morse theory for manifolds with boundary.
- **Research internship**, Institut Mathématique d'Orsay — Yves de Laszlo Mar–May 2025
  Co-authored a text on group representation theory from finite to solvable groups; proved foundational results including the Lie-Kolchin theorem.

Experience

- **Automatants**, CentraleSupélec student AI association Nov 2025–Apr 2026
  Guided deep learning projects (GAN for image generation, CNN for Alzheimer's detection, classifiers); brainstorming for AI hackathons and seminars, including geometric deep learning; educational slides for introductory ML sessions.
- **Data Analysis Internship**, TLD (ALVEST Group) Jun–Jul 2025
  Diagnosed failure causes from error logs of electric aircraft tractors. DBSCAN temporal clustering to separate root causes from secondary effects, then two composite scores — a cause/effect axis from a Beta-law fit of in-burst positions and a recurrent/impulsive axis from the dispersion of appearance days — with a rank-based metric for timestamp collisions, validated against hierarchical clustering. Delivered as an interactive Python/Dash tool for engineers.
- **Data Science Internship**, Johnson/Luke Jul–Aug 2024
  NLP and ML on customer feedback for a Lebanese bank's mobile app, in a context of eroded consumer confidence following the country's financial crisis; structured insights in a graph database to identify retention issues. Also contributed to UX testing of the Dubai Municipality website (25% improvement in user clarity).

Selected Projects (github.com/aelkas)

- **Shaaer — Arabic Poetry Transformer** (Python): GPT trained from scratch on classical Arabic poetry with a custom grapheme-cluster-aware BPE tokenizer; ~3.01 bits per character, below the bzip2 baseline.
- **Quad-trees** (C): binary images as quadtrees with morphological operations and a compression pass sharing structurally identical subtrees, turning the tree into a DAG.
- **Graphs and Logic** (Python): object-oriented directed open graphs extended into Boolean circuits and adders; gate transformations delegated to specializations of an abstract node class.
- **Colt Express** (Java) and **Maze generation and solving** (OCaml): full game implementations with graphical and animated interfaces.

Skills and Seminars

- **Programming:** Python, C, C++, Java, OCaml · **Databases:** PostgreSQL, Neo4j · **Tools:** LaTeX, Git
- **Languages:** French (fluent), English (TOEFL iBT 108/120), Arabic (fluent)
- **Seminars:** DataShape seminars, INRIA Saclay, on the recommendation of Nina Otter (2026); summer school on percolation under Thomas Hutchcroft, IHES (2025); Philosophy and Mathematics seminars, ENS PSL (2023–2024)
- **Interests:** computational topology and geometry, representation theory, geometric deep learning; violin, sports